

B4 Simultaneous equations

Name: _____

Class: _____

Date: _____

Time: **454 minutes**

Marks: **385 marks**

Comments:

Q1.

Point C has coordinates $(c, 2)$ and point D has coordinates $(6, d)$.

The line $y + 4x = 11$ is the perpendicular bisector of CD .

Find c and d .

(Total 5 marks)

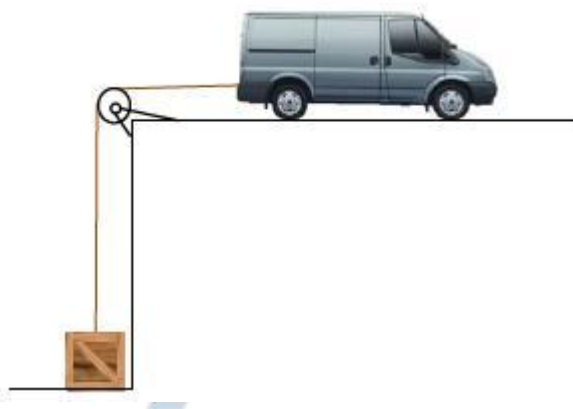
Q2.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A van of mass 1300 kg and a crate of mass 300 kg are connected by a light inextensible rope.

The rope passes over a light smooth pulley, as shown in the diagram.

The rope between the pulley and the van is horizontal.



Initially, the van is at rest and the crate rests on the lower level. The rope is taut.

The van moves away from the pulley to lift the crate from the lower level.

The van's engine produces a constant driving force of 5000 N .

A constant resistance force of magnitude 780 N acts on the van.

Assume there is no resistance force acting on the crate.

(a) (i) Draw a diagram to show the forces acting on the crate while it is being lifted.

(1)

(ii) Draw a diagram to show the forces acting on the van while the crate is being lifted.

(1)

(b) Show that the acceleration of the van is 0.80 m s^{-2}

(4)

(c) Find the tension in the rope.

(2)

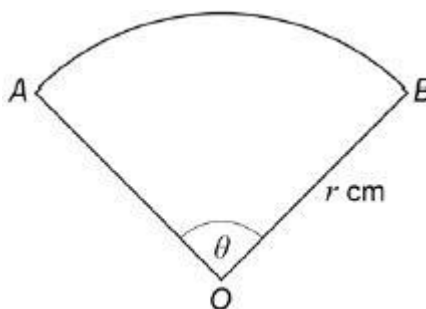
- (d) Suggest how the assumption of a constant resistance force could be refined to produce a better model.

(1)

(Total 9 marks)

Q3.

The diagram shows a sector AOB of a circle with centre O and radius r cm.



The angle AOB is θ radians.

The sector has area 9 cm^2 and perimeter 15 cm .

- (a) Show that r satisfies the equation $2r^2 - 15r + 18 = 0$

(4)

- (b) Find the value of θ . Explain why it is the only possible value.

(4)

(Total 8 marks)

Q4.

An open-topped fish tank is to be made for an aquarium.

It will have a square horizontal base, rectangular vertical sides and a volume of 60 m^3

The materials cost:

- £15 per m^2 for the base
- £8 per m^2 for the sides.

- (a) Modelling the sides and base of the fish tank as laminae, use calculus to find the height of the tank for which the overall cost of the materials has its minimum value.

Fully justify your answer.

(8)

- (b) (i) **In reality, the thickness of the base and sides of the tank is 2.5 cm**

Briefly explain how you would refine your modelling to take account of the thickness of the sides and base of the tank.

(1)

- (ii) How would your refinement affect your answer to part (a)?

(1)
(Total 10 marks)

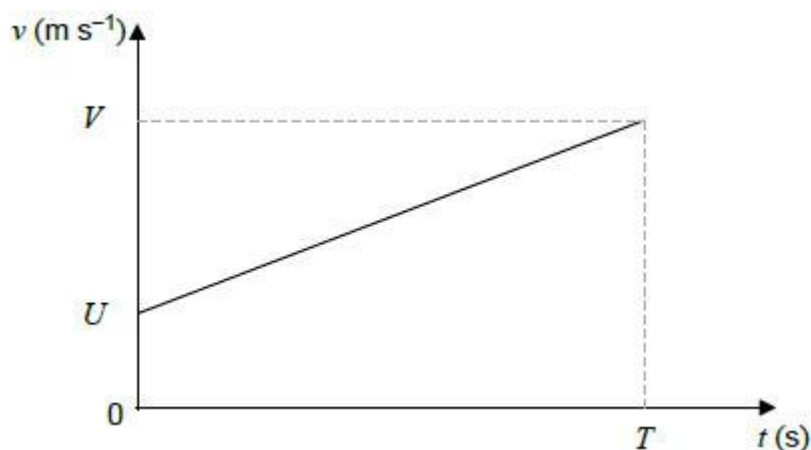
Q5.

A particle moves on a straight line with a constant acceleration, $a \text{ m s}^{-2}$.

The initial velocity of the particle is $U \text{ m s}^{-1}$

After T seconds the particle has velocity $V \text{ m s}^{-1}$

This information is shown on the velocity-time graph.



The displacement, S metres, of the particle from its initial position at time T seconds is given by the formula

$$S = \frac{1}{2}(U + V)T$$

- (a) By considering the gradient of the graph, or otherwise, write down a formula for a in terms of U , V and T .

(1)

- (b) Hence show that $V^2 = U^2 + 2aS$.

(3)

(Total 4 marks)

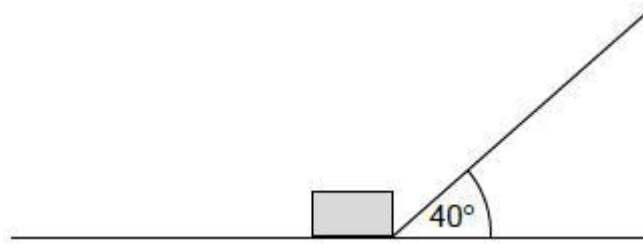
Q6.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg , being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons .



The coefficient of friction between the box and the board is μ

Model the box as a particle.

- (a) Show that $\mu = 0.83$

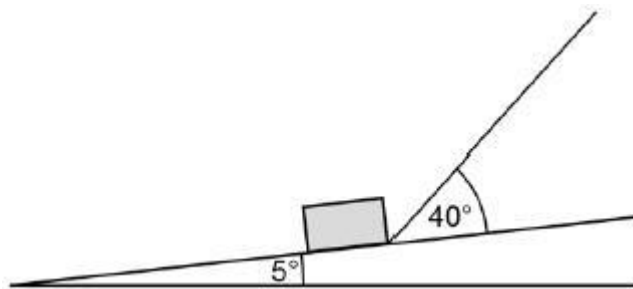
(4)

- (b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.

The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



- (i) Draw a diagram to show the forces acting on the box as it moves.

(1)

- (ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2} .

(7)

(Total 12 marks)

Q7.

The point A has coordinates $(-3, 2)$ and the point B has coordinates $(7, k)$.

The line AB has equation $3x + 5y = 1$.

- (a) (i) Show that $k = -4$.

(1)

- (ii) Hence find the coordinates of the midpoint of AB .

(2)

- (b) Find the gradient of AB . (2)
- (c) A line which passes through the point A is perpendicular to the line AB . Find an equation of this line, giving your answer in the form $px + qy + r = 0$, where p , q and r are integers. (3)
- (d) The line AB , with equation $3x + 5y = 1$, intersects the line $5x + 8y = 4$ at the point C . Find the coordinates of C . (3)
- (Total 11 marks)**

Q8.

A curve has equation $y = 2x^2 - x - 1$ and a line has equation $y = k(2x - 3)$, where k is a constant.

- (a) Show that the x -coordinate of any point of intersection of the curve and the line satisfies the equation

$$2x^2 - (2k + 1)x + 3k - 1 = 0$$

(1)

- (b) The curve and the line intersect at two distinct points.

- (i) Show that $4k^2 - 20k + 9 > 0$. (3)

- (ii) Find the possible values of k . (4)
- (Total 8 marks)**

Q9.

The line AB has equation $3x - 4y + 5 = 0$.

- (a) The point with coordinates $(p, p + 2)$ lies on the line AB . Find the value of the constant p . (2)
- (b) Find the gradient of AB . (2)
- (c) The point A has coordinates $(1, 2)$. The point $C(-5, k)$ is such that AC is perpendicular to AB . Find the value of k . (3)
- (d) The line AB intersects the line with equation $2x - 5y = 6$ at the point D . Find the coordinates of D .

(3)
(Total 10 marks)

Q10.

A curve has equation $y = x^5 - 2x^2 + 9$. The point P with coordinates $(-1, 6)$ lies on the curve.

- (a) Find the equation of the tangent to the curve at the point P , giving your answer in the form $y = mx + c$.

(5)

- (b) The point Q with coordinates $(2, k)$ lies on the curve.

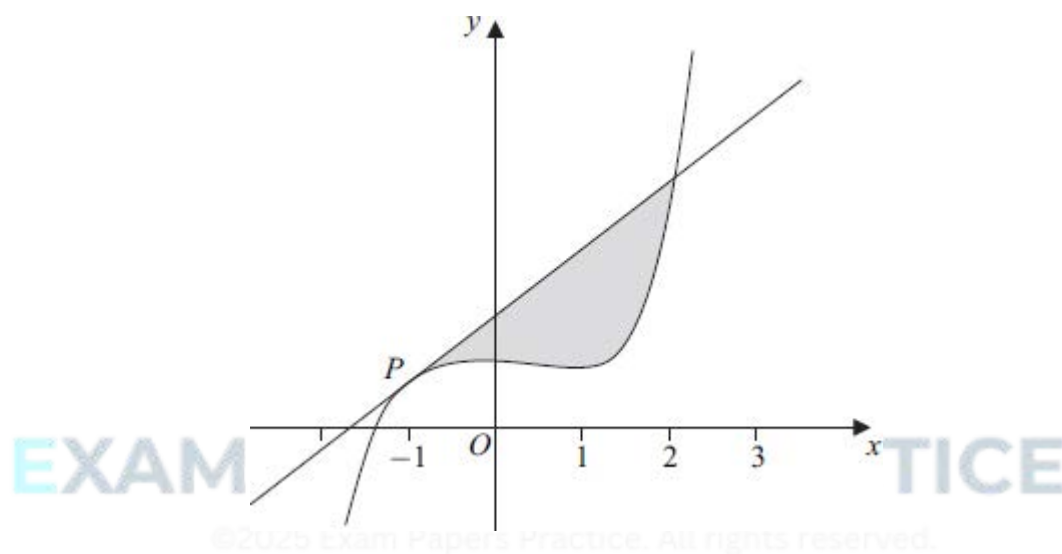
- (i) Find the value of k .

(1)

- (ii) Verify that Q also lies on the tangent to the curve at the point P .

(1)

- (c) The curve and the tangent to the curve at P are sketched below.



- (i) Find $\int_{-1}^2 (x^5 - 2x^2 + 9) dx$.

(5)

- (ii) Hence find the area of the shaded region bounded by the curve and the tangent to the curve at P .

(3)

(Total 15 marks)

Q11.

A circle with centre C has equation $x^2 + y^2 + 14x - 10y + 49 = 0$.

- (a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

- (b) Write down:

- (i) the coordinates of C ;

(3)

- (ii) the radius of the circle.

(2)

- (c) Sketch the circle.

(2)

- (d) A line has equation $y = kx + 6$, where k is a constant.

- (i) Show that the x -coordinates of any points of intersection of the line and the circle satisfy the equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$.

(2)

- (ii) The equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$ has equal roots. Show that

$$12k^2 - 7k - 12 = 0$$

(3)

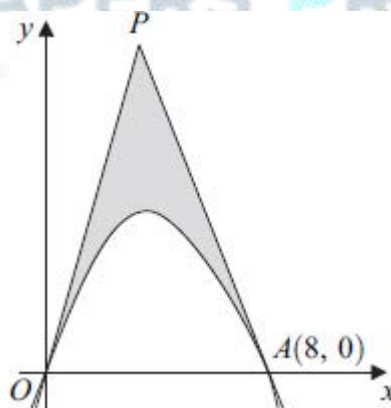
- (iii) Hence find the values of k for which the line is a tangent to the circle.

(2)

(Total 14 marks)

Q12.

The diagram shows part of a curve crossing the x -axis at the origin O and at the point $A(8, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve has equation

$$y = 12x - 3x^{\frac{5}{3}}$$

- (a) Find $\frac{dy}{dx}$. (2)
- (b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O . (2)
- (ii) Show that the equation of the tangent at $A(8, 0)$ is $y + 8x = 64$. (3)
- (c) Find $\int \left(12x - 3x^{\frac{5}{3}} \right) dx$. (3)
- (d) Calculate the area of the shaded region bounded by the curve from O to A and the tangents OP and AP . (7)
- (Total 17 marks)**

Q13.

The line AB has equation $4x - 3y = 7$.

- (a) (i) Find the gradient of AB . (2)
- (ii) Find an equation of the straight line that is parallel to AB and which passes through the point $C(3, -5)$, giving your answer in the form $px + qy = r$, where p , q and r are integers. (3)
- (b) The line AB intersects the line with equation $3x - 2y = 4$ at the point D . Find the coordinates of D . (3)
- (c) The point E with coordinates $(k - 2, 2k - 3)$ lies on the line AB . Find the value of the constant k . (2)
- (Total 10 marks)**

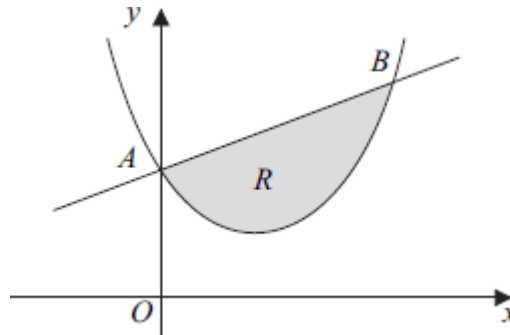
Q14.

- (a) (i) Express $x^2 - 3x + 5$ in the form $(x - p)^2 + q$. (2)
- (ii) Hence write down the equation of the line of symmetry of the curve with equation

$$y = x^2 - 3x + 5.$$

(1)

- (b) The curve C with equation $y = x^2 - 3x + 5$ and the straight line $y = x + 5$ intersect at the point $A(0, 5)$ and at the point B , as shown in the diagram below.



- (i) Find the coordinates of the point B .

(3)

- (ii) Find $\int (x^2 - 3x + 5) dx$.

(3)

- (iii) Find the area of the shaded region R bounded by the curve C and the line segment AB .

(4)

(Total 13 marks)

Q15.

The gradient, $\frac{dy}{dx}$, of a curve C at the point (x, y) is given by

$$\frac{dy}{dx} = 20x - 6x^2 - 16$$

- (a) (i) Show that y is increasing when $3x^2 - 10x + 8 < 0$.

(2)

- (ii) Solve the inequality $3x^2 - 10x + 8 < 0$.

(4)

- (b) The curve C passes through the point $P(2, 3)$.

- (i) Verify that the tangent to the curve at P is parallel to the x-axis.

(2)

- (ii) The point $Q(3, -1)$ also lies on the curve. The normal to the curve at Q and the tangent to the curve at P intersect at the point R . Find the coordinates of R .

(7)
 (Total 15 marks)

Q16.

The line AB has equation $3x + 2y = 7$. The point C has coordinates $(2, -7)$.

- (a) (i) Find the gradient of AB . (2)
- (ii) The line which passes through C and which is parallel to AB crosses the y -axis at the point D . Find the y -coordinate of D . (3)
- (b) The line with equation $y = 1 - 4x$ intersects the line AB at the point A . Find the coordinates of A . (3)
- (c) The point E has coordinates $(5, k)$. Given that CE has length 5, find the two possible values of the constant k . (3)
- (Total 11 marks)

Q17.

- (a) (i) Express $4 - 10x - x^2$ in the form $p - (x + q)^2$ (2)
- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = 4 - 10x - x^2$. (1)
- (b) The curve C has equation $y = 4 - 10x - x^2$ and the line L has equation $y = k(4x - 13)$, where k is a constant.
- (i) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation
- $$x^2 + 2(2k + 5)x - (13k + 4) = 0$$
- (1)
- (ii) Given that the curve C and the line L intersect in two distinct points, show that
- $$4k^2 + 33k + 29 > 0$$
- (3)
- (iii) Solve the inequality $4k^2 + 33k + 29 > 0$. (4)
- (Total 11 marks)

Q18.

The line AB has equation $7x + 3y = 13$.

- (a) Find the gradient of AB .

(2)

- (b) The point C has coordinates $(-1, 3)$.

- (i) Find an equation of the line which passes through the point C and which is parallel to AB .

(2)

- (ii) The point $\left(1\frac{1}{2}, -1\right)$ is the mid-point of AC . Find the coordinates of the point A .

(2)

- (c) The line AB intersects the line with equation $3x + 2y = 12$ at the point B . Find the coordinates of B .

(3)

(Total 9 marks)

Q19.

A circle has centre $C(3, -8)$ and radius 10.

- (a) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k$$

(2)

- (b) Find the x -coordinates of the points where the circle crosses the x -axis.

(3)

- (c) The tangent to the circle at the point A has gradient $\frac{5}{2}$. Find an equation of the line CA , giving your answer in the form $rx + sy + t = 0$, where r , s and t are integers.

(3)

- (d) The line with equation $y = 2x + 1$ intersects the circle.

- (i) Show that the x -coordinates of the points of intersection satisfy the equation

$$x^2 + 6x - 2 = 0$$

(3)

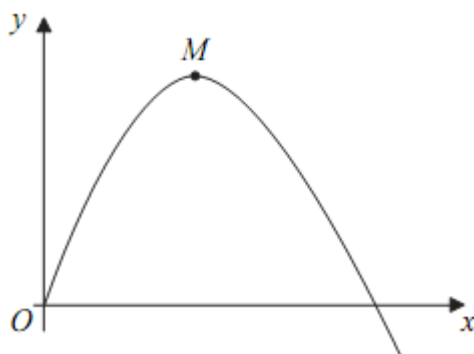
- (ii) Hence show that the x -coordinates of the points of intersection are of the form $m \pm \sqrt{n}$, where m and n are integers.

(2)

(Total 13 marks)

Q20.

The diagram shows part of a curve with a maximum point M .



The curve is defined for $x \geq 0$ by the equation

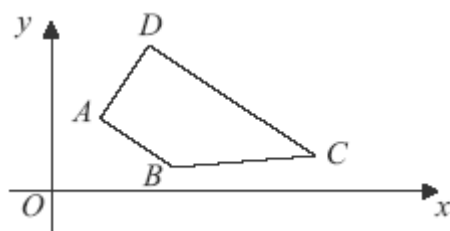
$$y = 6x - 2x^{\frac{3}{2}}$$

- (a) Find $\frac{dy}{dx}$. (3)
- (b) (i) Hence find the coordinates of the maximum point M . (3)
- (ii) Write down the equation of the normal to the curve at M . (1)
- (c) The point $P\left(\frac{9}{4}, \frac{27}{4}\right)$ lies on the curve. (2)
- (i) Find an equation of the normal to the curve at the point P , giving your answer in the form $ax + by = c$, where a , b and c are positive integers. (4)
- (ii) The normals to the curve at the points M and P intersect at the point R . Find the coordinates of R . (2)

(Total 13 marks)

Q21.

The trapezium $ABCD$ is shown below.



The line AB has equation $2x + 3y = 14$ and DC is parallel to AB .

- (a) Find the gradient of AB . (2)
- (b) The point D has coordinates $(3, 7)$.
- (i) Find an equation of the line DC . (2)
- (ii) The angle BAD is a right angle. Find an equation of the line AD , giving your answer in the form $mx + ny + p = 0$, where m , n and p are integers. (4)
- (c) The line BC has equation $5y - x = 6$. Find the coordinates of B . (3)
- (Total 11 marks)**

Q22.

The 25th term of an arithmetic series is 38.

The sum of the first 40 terms of the series is 1250.

- (a) Show that the common difference of this series is 1.5. (6)
- (b) Find the number of terms in the series which are less than 100. (3)
- (Total 9 marks)**

Q23.

The line AB has equation $3x + 5y = 11$.

- (a) (i) Find the gradient of AB . (2)
- (ii) The point A has coordinates $(2, 1)$. Find an equation of the line which passes through the point A and which is perpendicular to AB . (3)
- (b) The line AB intersects the line with equation $2x + 3y = 8$ at the point C . Find the

coordinates of C .

(3)

(Total 8 marks)

Q24.

The curve C has equation $y = k(x^2 + 3)$, where k is a constant.

The line L has equation $y = 2x + 2$.

- (a) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation

$$kx^2 - 2x + 3k - 2 = 0$$

(1)

- (b) The curve C and the line L intersect in two distinct points.

- (i) Show that

$$3k^2 - 2k - 1 < 0$$

(4)

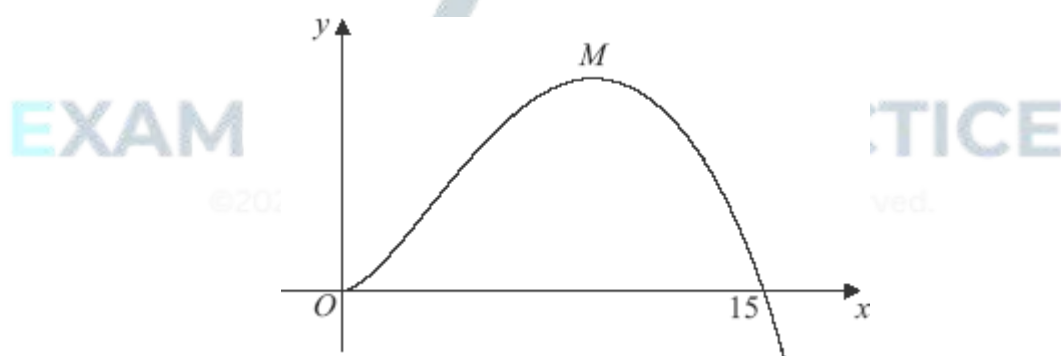
- (ii) Hence find the possible values of k .

(4)

(Total 9 marks)

Q25.

The diagram shows part of a curve with a maximum point M .



The equation of the curve is

$$y = 15x^{\frac{3}{2}} - x^{\frac{5}{2}}$$

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Hence find the coordinates of the maximum point M . (4)
- (c) The point $P(1, 14)$ lies on the curve. Show that the equation of the tangent to the curve at P is $y = 20x - 6$. (3)
- (d) The tangents to the curve at the points P and M intersect at the point R . Find the length of RM . (3)
- (Total 13 marks)**

Q26.

A circle with centre C has equation $x^2 + y^2 - 10y + 20 = 0$.

- (a) By completing the square, express this equation in the form

$$x^2 + (y - b)^2 = k$$

(2)

- (b) Write down:

- (i) the coordinates of C ; (1)
- (ii) the radius of the circle, leaving your answer in surd form. (1)

- (c) A line has equation $y = 2x$.

- (i) Show that the x -coordinate of any point of intersection of the line and the circle satisfies the equation $x^2 - 4x + 4 = 0$. (2)
- (ii) Hence show that the line is a tangent to the circle and find the coordinates of the point of contact, P . (3)

- (d) Prove that the point $Q(-1, 4)$ lies inside the circle. (2)

(Total 11 marks)

Q27.

The curve C has equation $y = x^2 + 7$. The line L has equation $y = k(3x + 1)$, where k is a constant.

- (a) Show that the x -coordinates of any points of intersection of the line L with the curve C satisfy the equation

$$x^2 - 3kx + 7 - k = 0$$

(1)

- (b) The curve C and the line L intersect in two distinct points. Show that

$$9k^2 + 4k - 28 > 0$$

(3)

- (c) Solve the inequality $9k^2 + 4k - 28 > 0$.

(4)

(Total 8 marks)

Q28.

The straight line L has equation $y = 3x - 1$ and the curve C has equation

$$y = (x + 3)(x - 1)$$

- (a) Sketch on the same axes the line L and the curve C , showing the values of the intercepts on the x -axis and the y -axis.

(5)

- (b) Show that the x -coordinates of the points of intersection of L and C satisfy the equation $x^2 - x - 2 = 0$.

(2)

- (c) Hence find the coordinates of the points of intersection of L and C .

(4)

(Total 11 marks)

Q29.

Two numbers, x and y , are such that $3x + y = 9$, where $x \geq 0$ and $y \geq 0$.

It is given that $V = xy^2$.

- (a) Show that $V = 81x - 54x^2 + 9x^3$.

(2)

- (b) (i) Show that $\frac{dV}{dx} = k(x^2 - 4x + 3)$, and state the value of the integer k .

(4)

- (ii) Hence find the two values of x for which $\frac{dV}{dx} = 0$.

(2)

- (c) Find $\frac{d^2 V}{dx^2}$. (2)

- (d) (i) Find the value of $\frac{d^2 V}{dx^2}$ for each of the two values of x found in part (b)(ii). (1)

- (ii) Hence determine the value of x for which V has a maximum value. (1)

- (iii) Find the maximum value of V . (1)

(Total 13 marks)

Q30.

The line AB has equation $3x + 5y = 8$ and the point A has coordinates $(6, -2)$.

- (a) (i) Find the gradient of AB . (2)

- (ii) Hence find an equation of the straight line which is perpendicular to AB and which passes through A . (3)

- (b) The line AB intersects the line with equation $2x + 3y = 3$ at the point B . Find the coordinates of B . (3)

- (c) The point C has coordinates $(2, k)$ and the distance from A to C is 5. Find the **two** possible values of the constant k . (3)

(Total 11 marks)

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Q31.

A circle with centre C has equation $x^2 + y^2 + 2x - 12y + 12 = 0$.

- (a) By completing the square, express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2 \quad (3)$$

- (b) Write down:

- (i) the coordinates of C ; (1)

- (ii) the radius of the circle. (1)
- (c) Show that the circle does **not** intersect the x -axis. (2)
- (d) The line with equation $x + y = 4$ intersects the circle at the points P and Q .
- (i) Show that the x -coordinates of P and Q satisfy the equation
- $$x^2 + 3x - 10 = 0$$
- (3)
- (ii) Given that P has coordinates $(2, 2)$, find the coordinates of Q . (2)
- (iii) Hence find the coordinates of the midpoint of PQ . (2)
- (Total 14 marks)

Q32.

- (a) (i) Express $x^2 + 10x + 19$ in the form $(x + p)^2 + q$, where p and q are integers. (2)
- (ii) Write down the coordinates of the vertex (minimum point) of the curve with equation $y = x^2 + 10x + 19$. (2)
- (iii) Write down the equation of the line of symmetry of the curve $y = x^2 + 10x + 19$. (1)
- (iv) Describe geometrically the transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 10x + 19$. (3)
- (b) Determine the coordinates of the points of intersection of the line $y = x + 11$ and the curve $y = x^2 + 10x + 19$. (4)
- (Total 12 marks)

Q33.

- (a) (i) Express $x^2 - 4x + 9$ in the form $(x - p)^2 + q$, where p and q are integers. (2)
- (ii) Hence, or otherwise, state the coordinates of the minimum point of the curve with equation $y = x^2 - 4x + 9$. (2)

(b) The line L has equation $y + 2x = 12$ and the curve C has equation $y = x^2 - 4x + 9$.

(i) Show that the x -coordinates of the points of intersection of L and C satisfy the equation

$$x^2 - 2x - 3 = 0$$

(1)

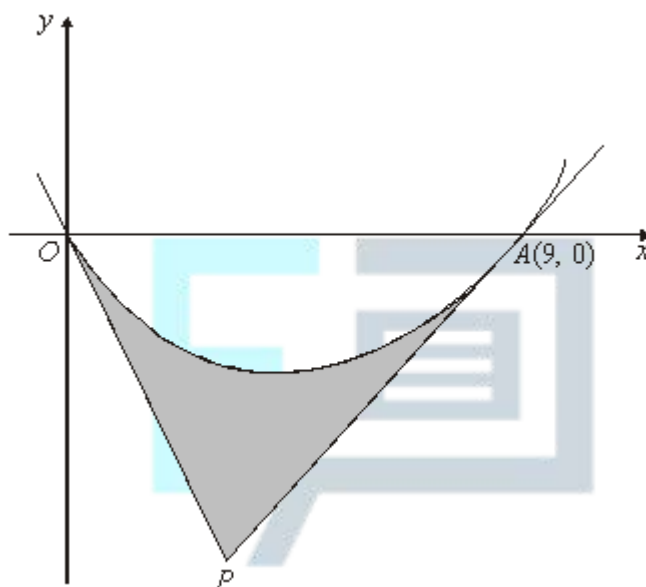
(ii) Hence find the coordinates of the points of intersection of L and C .

(4)

(Total 9 marks)

Q34.

A curve, drawn from the origin O , crosses the x -axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O .

(2)

(ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$.

(3)

(iii) Hence find the coordinates of the point P where the two tangents meet.

(3)

(c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$.

(3)

(d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP .

(5)

(Total 18 marks)

Q35.

The point A has coordinates $(1, 7)$ and the point B has coordinates $(5, 1)$.

(a) (i) Find the gradient of the line AB .

(2)

(ii) Hence, or otherwise, show that the line AB has equation $3x + 2y = 17$.

(2)

(b) The line AB intersects the line with equation $x - 4y = 8$ at the point C . Find the coordinates of C .

(3)

(c) Find an equation of the line through A which is perpendicular to AB .

(3)

(Total 10 marks)

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